Decidability 05

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An Undecidable Language

Theorem 4.11

Let $A_{\text{TM}} = \{\langle M, w\rangle \mid M \text{ is a TM and } M \text{ accepts } w\}$. A_{TM} is an undecidable language.

- Assume that A_{TM} is decidable
- Thus, there exists a Turing machine (decider) H that decides A_{TM}
  - Given a TM M and a string w in the form $\langle M, w\rangle$, H can decide whether $\langle M, w\rangle \in A_{\text{TM}}$ or not
- Let $H(\langle M, w\rangle)$ represents running TM H on input $\langle M, w\rangle$
- Since H is a decider for A_{TM}, we have

$$H(\langle M, w\rangle) = \begin{cases} accept & \text{if } \langle M, w\rangle \in A_{\text{TM}} \rightsquigarrow M \text{ accepts } w \\ reject & \text{if } \langle M, w\rangle \notin A_{\text{TM}} \rightsquigarrow M \text{ does not accept } w \end{cases}$$

  where M is a TM and w is a string
- Note that H always halts since it is a decider

An Undecidable Language

- Note that an encoding of a machine M denoted by $\langle M \rangle$ is simply a string
  - Therefore, we can run M on input $\langle M \rangle$ if we want to
- Suppose we want to know whether M accepts $\langle M \rangle$
  - We cannot simply run M on input $\langle M \rangle$
    - M may loop indefinitely on input $\langle M \rangle$ and we may not know
  - But now we have TM H (a decider for A_{TM}
    - Simply run H on input $\langle M, \langle M \rangle \rangle$
    - If H accepts $\langle M, \langle M \rangle \rangle$, M accepts $\langle M \rangle$
    - If H rejects $\langle M, \langle M \rangle \rangle$, M does not accept $\langle M \rangle$

An Undecidable Language

- Now, create a machine D as follows:

  $D =$ "On input $\langle M \rangle$, where M is a TM:

  1. Run H on input $\langle M, \langle M \rangle \rangle$.
  2. Output the opposite of what H outputs. That is, if H accepts, $reject$; and if H rejects, $accept$."

- That is

$$D(\langle M \rangle) = \begin{cases} accept & \text{if } H(\langle M, \langle M \rangle \rangle) = reject \rightsquigarrow M \text{ does not accept } \langle M \rangle \\ reject & \text{if } H(\langle M, \langle M \rangle \rangle) = accept \rightsquigarrow M \text{ accepts } \langle M \rangle \end{cases}$$

An Undecidable Language

- Note that an encoding of a machine D denoted by $\langle D \rangle$ is also a string
  - Therefore, we can run D on input $\langle D \rangle$
- Let's substitute M by D in the definite of TM D on the previous slide

$$D(\langle D \rangle) = \begin{cases} accept & \text{if } H(\langle D, \langle D \rangle \rangle) = reject \rightsquigarrow D \text{ does not accept } \langle D \rangle \\ reject & \text{if } H(\langle D, \langle D \rangle \rangle) = accept \rightsquigarrow D \text{ accepts } \langle D \rangle \end{cases}$$

- The above equation says the following:
  - TM D accepts $\langle D \rangle$ if D does not accept $\langle D \rangle$
  - TM D rejects $\langle D \rangle$ if D accepts $\langle D \rangle$

  which is contradict to itself
- Therefore, A_{TM} is undecidable

Unrecognizable Language

- Consider a **recognizable** language A
  - Since A is recognizable, there exists a machine M_A that accepts every string $s \in A$
  - If $s \notin A$, M_A may reject or loop indefinitely on input s
- Consider a **recognizable** language $\overline{A}$ (the complement of A)
  - Since $\overline{A}$ is recognizable, there exists a machine $M_{\overline{A}}$ that accepts every string $s \in \overline{A}$
  - If $s \notin \overline{A}$, $M_{\overline{A}}$ may reject or loop indefinitely on input s
- Both A and $\overline{A}$ are recognizable, is A decidable?
- To show that A is decidable, we have to perform the following:
  - Construct a TM M' that we think it is a decider for A
  - Show that M' accepts all strings in A
  - Show that M' rejects all strings not in A

Unrecognizable Language

- The following is an example of the TM M':

  $M' =$ "On input s:

  1. Run both M_A and $M_{\overline{A}}$ on input s in parallel.
  2. If M_A accepts s, *accept*; if $M_{\overline{A}}$ accepts s, *reject*."

- Case 1: Assume that $s \in A$. Since $s \in A$ and M_A recognizes A, by running M_A on input s, M_A will accept s. Since $s \notin \overline{A}$ and $M_{\overline{A}}$ recognizes $\overline{A}$, by running $M_{\overline{A}}$ on input s, $M_{\overline{A}}$ will not accept s. Since M_A accepts s, M' accepts s.
- Case 2: Assume that $s \notin A$. Since $s \notin A$ and M_A recognizes A, by running M_A on input s, M_A will not accept s. Since $s \in \overline{A}$ and $M_{\overline{A}}$ recognizes $\overline{A}$, by running $M_{\overline{A}}$ on input s, $M_{\overline{A}}$ will accept s. Since $M_{\overline{A}}$ accepts s, M' rejects s.
- This show that M' is a decider for A
- Therefore, A is decidable

Unrecognizable Language

- Now we just prove the following statement:

  If a language A is recognizable and its complement $\overline{A}$ is recognizable, A is decidable

- Let
  - $P \equiv A$ is recognizable
  - $Q \equiv \overline{A}$ is recognizable
  - $R \equiv A$ is decidable
- The above statement is simply $(P \wedge Q) \rightarrow R$
- From boolean algebra:

$$\begin{aligned}(P \wedge Q) \rightarrow R &\equiv \neg(P \wedge Q) \vee R \\ &\equiv (\neg P \vee \neg Q) \vee R \\ &\equiv (\neg P \vee R) \vee \neg Q \\ &\equiv \neg(P \wedge \neg R) \vee \neg Q \\ &\equiv (P \wedge \neg R) \rightarrow \neg Q\end{aligned}$$

Unrecognizable Language

- From previous slide, we have

$$(P \wedge Q) \rightarrow R \equiv (P \wedge \neg R) \rightarrow \neg Q$$

- By replacing P, Q, and R on the right side, we have the following statement:

  If A is recognizable and A is undecidable, $\overline{A}$ is unrecognizable
- Recall the following:
  - A_{TM} is recognizable
  - A_{TM} is undecidable
- From the above statement, $\overline{A_{\mathrm{TM}}}$ is unrecognizable

Conclusions

- To show that a language A is decidable:
  - Construct a TM M' that you think will decide A
  - Show that M' accepts all strings in A
  - Show that M' rejects all strings not in A
- The following TMs can be used as helper machines:
  - TM M that decides A_{DFA}

    $$A_{\text{DFA}} = \{\langle B, w\rangle \mid B \text{ is a DFA that accepts } w\}$$

  - TM N that decides A_{NFA}

    $$A_{\text{NFA}} = \{\langle B, w\rangle \mid B \text{ is a NFA that accepts } w\}$$

  - TM P that decides A_{REX}

    $$A_{\text{REX}} = \{\langle R, w\rangle \mid R \text{ is a regular expression that generates } w\}$$

  - TM T that decides E_{DFA}

    $$E_{\text{DFA}} = \{\langle B\rangle \mid B \text{ is a DFA and } L(B) = \emptyset\}$$

  - TM F that decides EQ_{DFA}

    $$EQ_{\text{DFA}} = \{\langle A, B\rangle \mid A \text{ and } B \text{ are DFAs and } L(A) = L(B)\}$$

Conclusions

- To show that two sets have the same size, simply show a correspondence between two sets
- A correspondence must be one-to-one and onto
  - Can be a definition of a function
  - Can be represented by a table (skip duplicate)
- To show that an infinite set is countable, simply show a correspondence with the set of natural number $\mathbb{N}$
- To show that an infinite set is uncountable:
  - Diagonalization: if all elements in the set is somehow infinite
  - Show a correspondence with a known uncountable set
- $A_{\text{TM}} = \{\langle M, w\rangle \mid M \text{ is a TM that accepts } w\}$ is undecidable
  - Given a TM M and a string w, there is no algorithm that can check whether M accepts w